

26.03.10

$$a^* = \arg \max_a Q_w(s, a)$$

Policy Gradient $\pi_\theta(a|s)$: Dist. on A .

$$J(\theta) = \mathbb{E}_{\pi_\theta} \left[\sum_{t=1}^T r^{t+1} \cdot r_t \right] \quad T=1$$

$$J(\theta) = \sum_s d(s) \cdot \sum_a \pi_\theta(a|s) \cdot r_s^a$$

$$\nabla_\theta J(\theta) = \sum_s d(s) \sum_a \nabla_\theta \pi_\theta(a|s) r_s^a$$

$$= \sum_s d(s) \sum_a \pi_\theta(a|s) \cdot \frac{\nabla_\theta \pi_\theta(a|s)}{\pi_\theta(a|s)} \cdot r_s^a$$

$$= \sum_s d(s) \sum_a \pi_\theta(a|s) \cdot \nabla_\theta \log \pi_\theta(a|s) \cdot r_s^a$$

$$= \mathbb{E}_{\pi_\theta} \left[\underbrace{\nabla_\theta \log \pi_\theta}_{\text{Score}} \cdot r \right]$$